Weekly Meeting

25.08.07

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Experiment Setting

- Dataset
- Alpaca Eval QA set (805)
  a. GPT-4o-mini (w. temperature 0.7)
  b. Llama-3.1-8B-Inst (w. temperature 1.0)x2
  c. Llama-3.1-8B-Inst (w. greedy decoding)

  *max_length = 512

- Pair-wise test
  - GPT-4o-mini (w. greedy decoding)

Experiment Setting

- Dataset
- Total:
  - 1416 pair-wise dataset
  - 858 train / 284 valid / 284 test

Experiment Setting

- Preprocess
  - Delete cycle (a>b>c>a)
  - Get two dataset for each query (a>b, a>c)
  - Response token → get final last hidden layer for each token(R^{4096})
  - Next token embedding* → (R^{4096})

Experiment Setting

- Input Format
  - JSONL row
    - { "q": prompt, "win": preferred answer, "lose": dispreferred answer }
  - Token limits : q ≤ 128 tokens, win/lose ≤ 512 tokens
  - For each time-step t (1…512)
    - sequence = (q_ids + answer_ids[:t+1])[-512:]

- Loss : BCEWithLogits(s_win − s_lose , 1)
- Accuracy : mean(s_win > s_lose)

Experiment Setting

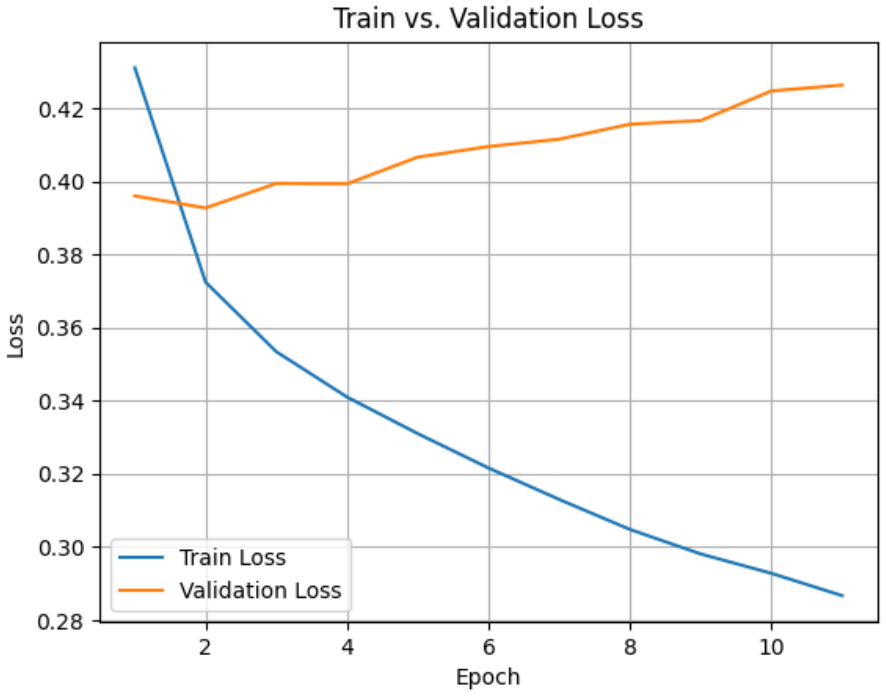
- Hyper-Parameters
  - Optimizer: AdamW
  - Learning rate: 1×10^{-4}
  - Weight decay: 1×10^{-4}
  - Epochs: 32
  - Max sequence length: unlimited
  - Dropout: 0.25
  - Hidden width: 2, 4, 8, 16, 32

- Input dimension: 4096 (only hidden layer)

Experiment Setting

- Results
  - Not generalizing
  - Overfitting detected
  - Similar pattern regardless of width
  - Accuracy ≈ 62%

```
ep10: 96it [00:12,  7.51it/s]
ep10  train 0.2922 | val 0.4247  acc 62.270%
ep11: 96it [00:12,  7.62it/s]
ep11  train 0.2867 | val 0.4263  acc 62.531%
▮ Early stopping triggered
[TEST] loss 0.4179 | acc 62.092%
```



Experiment Setting

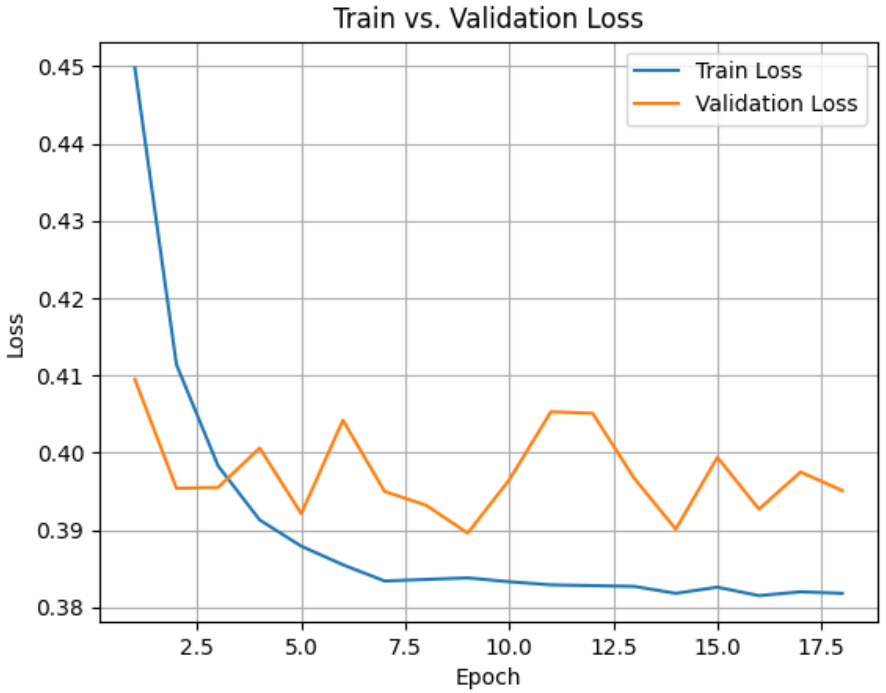
- Hyper-Parameters
  - Optimizer: AdamW, SGD
  - Momentum: 0.9(SGD)
  - Learning rate: 1×10^{-4}
  - Weight decay: 1×10^{-4}
  - Epochs: 32
  - Max sequence length: unlimited
  - Dropout: 0.25
  - Hidden width: 1, 2, 4, 8, 16, 32

- Input dimension: 8192 (hidden layer + embedding)

Experiment Setting

- Results(AdamW)
  - Not generalizing
  - Overfitting detected
  - Similar pattern regardless of width
  - Accuracy ≈ 62%

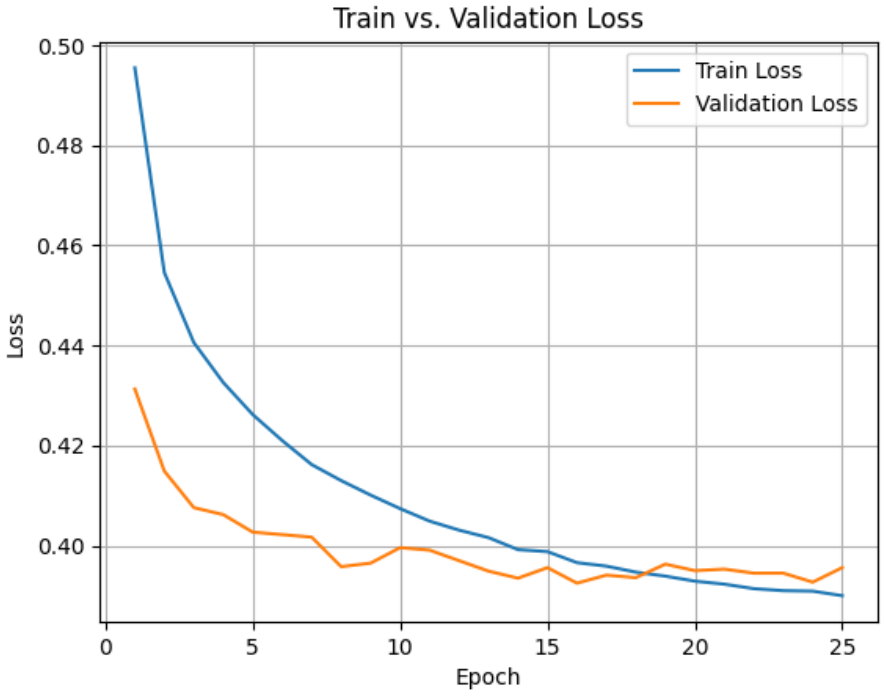
```
ep16: 164it [00:38,  4.21it/s]
ep16  train 0.3815 | val 0.3927  acc 65.598%
ep17: 164it [00:38,  4.24it/s]
ep17  train 0.3820 | val 0.3975  acc 65.303%
ep18: 164it [00:39,  4.12it/s]
ep18  train 0.3818 | val 0.3951  acc 65.389%
▌ Early stopping triggered
[TEST] loss 0.4093 | acc 62.787%
```



Experiment Setting

- Results(SGD, width=1)
  - Not generalizing
  - Overfitting detected
  - Similar pattern regardless of width
  - Accuracy ≈ 62%

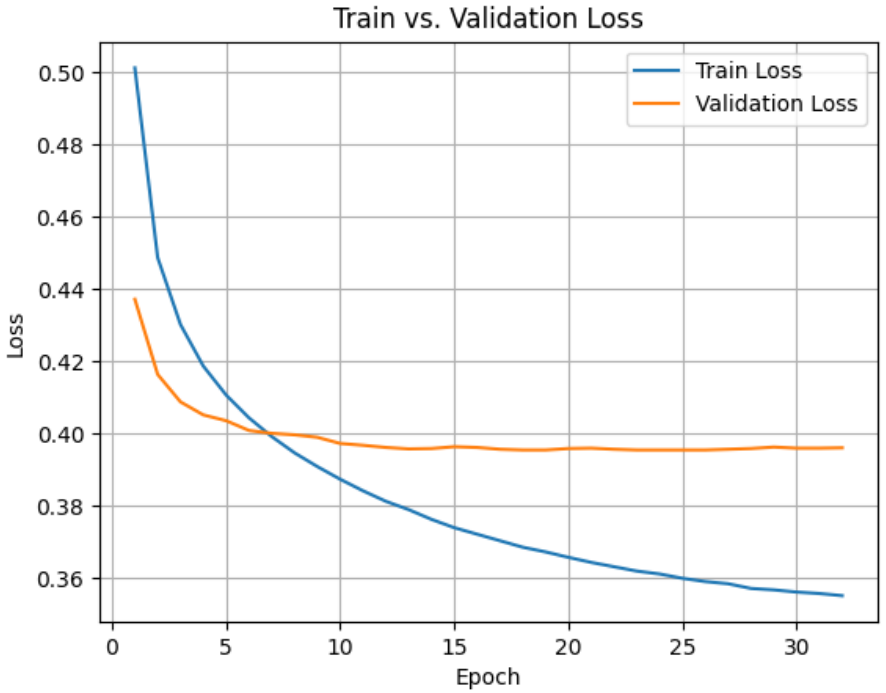
```
ep23: 164it [00:37,  4.42it/s]
ep23  train 0.3910 | val 0.3945  acc 65.464%
ep24: 164it [00:38,  4.26it/s]
ep24  train 0.3909 | val 0.3927  acc 65.507%
ep25: 164it [00:36,  4.46it/s]
ep25  train 0.3900 | val 0.3956  acc 65.425%
▌ Early stopping triggered
[TEST] loss 0.4110 | acc 62.752%
```



Experiment Setting

- Results(SGD, width=8)
  - Not generalizing
  - Overfitting detected
  - Similar pattern regardless of width
  - Accuracy ≈ 62%

```
ep30  train 0.3562 | val 0.3960  acc 65.468%
ep31: 164it [00:38,  4.30it/s]
ep31  train 0.3558 | val 0.3960  acc 65.507%
ep32: 164it [00:38,  4.21it/s]
ep32  train 0.3552 | val 0.3961  acc 65.478%
▌ Early stopping triggered
[TEST] loss 0.4152 | acc 62.699%
```



TODO

- Discussion
  - Increased accuracy: data preprocessing
  - Increasing dataset size
  - Other loss function

- Generation & pair-wise test
  - Generate w. trained model
  - Pair-wise test w. greedy decoding response